Glossary: Network Protocols

Welcome! This alphabetized glossary contains many of the terms in this course. This comprehensive glossary also includes additional industry-recognized terms not used in course videos. These terms are important for you to recognize when working in the industry, participating in user groups, and participating in other certificate programs.

Term	Definition
Acknowledgment number (ACK number)	A number in TCP that indicates the next expected packet in the sequence, acknowledging the receipt of previous packets.
Data exfiltration	Unauthorized transfer of sensitive data from an organization to an external location is often detected through unusual data transfer patterns.
DHCP acknowledgment (ACK)	The message sent by the DHCP server confirms that an IP address has been leased to the requesting device.
DHCP discover	The initial message is sent by a device to locate a DHCP server on the network.
DHCP handshake	The sequence of four messages (Discover, Offer, Request, Acknowledgment) exchanged between a device and a DHCP server to obtain an IP address.
Domain name system (DNS)	A network service that translates domain names into IP addresses.
Dynamic host configuration protocol (DHCP)	A protocol that automatically assigns IP addresses to devices from a pool managed by a DHCP server, allowing them to connect to the network.
Egressing	The process of data leaving an interface.
Hypertext transfer protocol (HTTP)	A protocol used for browsing the web, operating on a request-response cycle.
Hypertext transfer protocol secure (HTTPS)	A secure version of HTTP that uses SSL certificates to encrypt data for secure communication over the internet.
Internet control message protocol (ICMP)	A network protocol is used for error messages and operational information.
Internet protocol television (IPTV)	A service that streams TV signals using both UDP and TCP, depending on the service and traffic direction.
Media access control address (MAC address)	A unique identifier is assigned to network interfaces for communications on the physical network segment. Each network frame contains a destination MAC address.
Quality of service (QoS)	Information related to network service quality, including service type and protocol usage in network flows.
Remote switch port analyzer (RSPAN)	A type of port mirroring that allows the destination port to be on a different switch from the one where the traffic is being collected.
Simple mail transfer protocol (SMTP)	A protocol for sending emails.
Simple network management protocol (SNMP)	A protocol for monitoring and managing network devices and can use both UDP and TCP.
Syslog	A standard protocol for message logging that separates message generation, storage, and analysis, providing a structured format for messages.
Transmission control protocol (TCP)	A connection-oriented protocol that ensures reliable and ordered data delivery between applications, using a three-way handshake for connection establishment.
Trivial file transfer protocol (TFTP)	A simple file transfer protocol using UDP for transferring very small files without connection overhead.
User datagram protocol (UDP)	A connectionless protocol that allows data packets to be sent without establishing a connection, offering faster communication.
Voice-over IP (VoIP)	A protocol for transmitting voice over the Internet, typically using UDP, but can also use TCP.
Web traffic	Network traffic related to web browsing and HTTP protocols is often a major part of overall network traffic.
Wireshark	A network protocol analyzer that is used to capture and examine network traffic. It can put a NIC into promiscuous mode to capture all packets on an interface.

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